

National University of Computer and Emerging Sciences
Islamabad Campus

Computer Architecture (EE3009)

Sessional-I Exam

Course Instructor(s):
Ms. Noor Ul Ain, Mr. Aqib Rehman, Mr. Shams Farooq
Section(s): All CS Sections

Total Time (Hrs): 1
Total Marks: 60
Total Questions: 3

Date: Feb 24, 2026

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

	Q-1	Q-2	Q-3	TOTAL
Marks Obtained				
Total	20	20	20	60

INSTRUCTIONS

- Final answers must be written in permanent pen; pencil answers will not be accepted.
- Solve on question paper, use extra sheet only if required.
- Only one extra sheet is allowed for rough work; please ensure it is stapled. any work done on the rough sheet will not be accepted for marking.
- Overwriting and cutting in the final answer on the question paper will result in a zero score and cannot be claimed for rechecking.
- Sharing calculators or any other tools is prohibited.
- Your data memory begins at address **0x00000100H**, and the stack starts at address **0x00000600H**.

[CLO 1: Understand why performance matters in computer design and identify key performance metrics such as *response time* and *throughput*]

[CLO 2: Understanding **RISC-V ISA**, interpret assembly and machine instructions, and write and analyze basic RISC-V programs.]

CLO 1:

Question 1

Marks 5+5+5+5=20

- I. A program consists of **5,000** floating point instructions and **25,000** integer instructions. Processor A has a clock rate of **2.0 GHz**. Floating point instructions take **7 cycles** and integer instructions take **1 cycle**. How long does it take for this processor to run the program?

Calculate Total Cycles

Floating-point cycles

= 5,000 × 7

= **35,000 cycles**

Integer cycles

= 25,000 × 1

= **25,000 cycles**

Total cycles

= 35,000 + 25,000

= **60,000 cycles**

Execution time = 60000/2.0x10⁹

Execution Time= 3.0 x 10⁻⁵

30 microsecond(μsec)

0.00003 sec

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II. What is the **average CPI** for this processor for the given program?

Average CPI=60,000/30,000
Avg Cpi=2

III. Processor A runs Program 2 consisting of **100,000** floating point instructions and **50,000** integer instructions. What is the **average CPI** for this program?

Total instruction= 100,000+50,000=150,000 instructions

Total Clock Cycles

Floating-point instructions: $100,000 \times 7 = 700,000$ cycles

Integer instructions: $50,000 \times 1 = 50,000$ cycles

Total cycles: $700,000 + 50,000 = 750,000$ cycles

Average CPI=750,000/150000 =5 Cpi Program 2

IV. Processor B has an average CPI for Program 2 of **3.5**. Its clock rate is **1.8 GHz**. How much time does it take to execute the program?

Total cycles = $150,000 \times 3.5 = 525,000$

Clock rate= 1.8 GHz= 1.8×10^9

Time= $525000 / 1.8 \times 10^9 = 2.92 \times 10^{-4} \text{ sec} = 0.000292 \text{ seconds}$

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CLO 2:

Question 2

Marks 5+5+5+5=20

- I. Given the following **C code**, write the equivalent **RISC-V** assembly code. Also write the output for C code. Cout is for your understanding no need to convert that to RISC instruction

C Code	RISC- V Code																				
<pre>// j=s1 and i=s0 int j=50; while(j>1) { for(int i=1;i<21;i=i*3) { cout<<i<<" "<<j<<"\n"; } j=j/4; }</pre> <table border="1" style="margin-top: 10px;"> <thead> <tr> <th colspan="2">Output</th> </tr> </thead> <tbody> <tr><td>1</td><td>50</td></tr> <tr><td>3</td><td>50</td></tr> <tr><td>9</td><td>50</td></tr> <tr><td>1</td><td>12</td></tr> <tr><td>3</td><td>12</td></tr> <tr><td>9</td><td>12</td></tr> <tr><td>1</td><td>3</td></tr> <tr><td>3</td><td>3</td></tr> <tr><td>9</td><td>3</td></tr> </tbody> </table>	Output		1	50	3	50	9	50	1	12	3	12	9	12	1	3	3	3	9	3	<pre># j=s1 and i=s0</pre>
Output																					
1	50																				
3	50																				
9	50																				
1	12																				
3	12																				
9	12																				
1	3																				
3	3																				
9	3																				

- II. Convert the following **RISC-V** to the equivalent **C code**.

RISC- V Code	C Code
<pre>#s0=could be Any +ve integer addi t0,zero,5 addi t1,zero,200 addi t2,zero,299 beq s0,t2,skip1 blt s0,t0,skip2 bgt t0,s0,skip2 skip1: add a0,zero,zero j skip3 skip2: addi a0,zero,1 skip3:</pre>	<pre>#s0 =x, a0= a</pre>

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III. Consider the following **RISC-V code**. For each instruction, update the contents of all affected registers in hexadecimal.

RISC- V Code	Registers Affected	
addi t0,zero,4 addi t1,zero,0x780 addi t2,zero,2 addi s0,zero,660 #decimal mul s0,s0,t2 div s0,s0,t0 xori s0,s0,-1 sra s0,s0,t1	S0 addi	0x00000294
	S0 mul	0x00000528
	S0 div	0x0000014a
	S0 xori	0xffffffffeb5
	S0 sra	0xffffffffeb5

IV. Write a **RISC V code** to load registers **s0=012345789** and **s1=0x7FFF856**. Also perform **1's complement** of the value stored in **s0** after the load. **Note:** Do not use any **pseudo instruction** like **li** or **not**

```

lui s0,0x12345          lui s0, 0x0075bc
addi s0,s0,0x789        addi s0,3304 #9xd15
xori s0,s0,-1
lui s1,0x80000
addi s1,s1,-1962
  
```

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V.

VI.

CLO 2:

Question 3

Marks 10+10=20

- I. Consider the following **RISC-V** program. For each iteration, record the changes made to memory given memory. After every iteration of the loop, the updated memory in **Hexa-decimal**. Also write a **short** analysis for the code is doing.

.data

```
ary1: .word 1,2,3,4
ary1size: .byte 0
```

main:

```
    la a0,ary1      #load address of ary1 to s0 register
    la a1,ary1size
    addi t0,t0,123   # very important value
    addi s1,s1,456   #very important value
    sub a1,a1,a0
    srli a1,a1,2
    addi t0,zero,0
```

for:

```
    bge t0,a1,exit
    lw s1,0(a0)
    slli s1,s1,4
    sw s1,0(a0)
    addi a0,a0,4
    addi t0,t0,1
    j for
```

exit:

```
    add t0,t0,zero
```

Memory

	0	1	2	3	4	5	6	7	8
000	0x10	0x20	0x30	0x40					

Short Answer:

Multiplying each element by 16

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II. The program above has been converted to a function call. You must complete the code by saving preserve and non-preserved registers by filling given empty spaces. following table could be a help

Preserved (callee-saved)	Nonpreserved (caller-saved)
Saved registers: s0-s11	Temporary registers: t0-t6
Return address: ra	Argument registers: a0-a7
Stack pointer: sp	

.data

```
ary1: .word 1,2,3,4
ary1size: .byte 0
```

```
main:
    la a0,ary1
    #load address of ary1 to
    #s0 register

    la a1,ary1size
    addi t0,t0,12
    # very important value
    addi s1,s1,456
    #very important value
    sub a1,a1,a0
    srli a1,a1,2
    addi sp,sp,-4
    sw t0,0(sp)

    jal ra,funct

    lw t0,0(sp)
    addi sp,sp,4

    add t0,t0,zero

    li a7, 10#terminating instructions
    ecall    #terminating instructions
```

```
funct:

    addi sp,sp,-8
    sw s1,0(sp)
    sw ra,4(sp)

    addi t0,zero,0
    for:
    bge t0,a1,exit
    lw s1,0(a0)
    slli s1,s1,4
    sw s1,0(a0)
    addi a0,a0,4
    addi t0,t0,1
    jal ra,funct2
    j for
    exit:
    lw ra,4(sp)
    lw s1,0(sp)
    addi sp,sp,8

    jr ra
    #####
    funct2:
        add zero,zero,zero
    jr ra
```

BEST OF LUCK